1. What is the primary reason LLMs don’t rely on full “words”?

A. Words are too short
B. Languages differ in spacing rules
C. Words are computationally expensive
D. Transformers cannot store word embeddings
Answer: B
Explanation: Many languages (German, Chinese) don’t use spaces consistently, so “words” are unreliable .

---

2. Subword tokenization mainly solves which problem?

A. Overfitting
B. Out-of-vocabulary words
C. Slow inference
D. Large embeddings
Answer: B
Explanation: Vocabulary grows endlessly; subwords prevent unknown tokens .

---

3. Why are Unicode characters alone too small as tokens?

A. They cannot represent numbers
B. They lose semantic meaning
C. They overload vector space
D. They increase model size
Answer: B
Explanation: Characters don’t capture meaningful units like morphemes .

---

4. Embeddings place tokens close together based on:

A. Spelling similarity
B. Random distribution

C. Semantic similarity
D. Frequency in dataset
Answer: C
Explanation: Vector space organizes meaning mathematically .

---

5. Why did RNNs fail for large-scale NLP?

A. They lack embeddings
B. Vanishing gradients + slow sequential processing
C. They require GPUs
D. They don't support characters
Answer: B
Explanation: They cannot maintain long-range context and cannot parallelize .

---

6. Positional encoding is needed because transformers:

A. Remove punctuation
B. Don't process tokens in order
C. Merge tokens incorrectly
D. Forget embeddings
Answer: B
Explanation: Transformers process all tokens in parallel and need explicit position info .

---

7. The Q vector in attention represents:

A. What information a token stores
B. The numerical position
C. What the token is searching for
D. The final output
Answer: C
Explanation: Q = query = what this token wants .

---

8. A transformer decoder masks future tokens because:

A. It reduces hallucination
B. It enforces autoregressive generation
C. It shortens context
D. It improves training speed
Answer: B
Explanation: Decoder should only see previous tokens when predicting next ones .

---

9. Multi-head attention improves performance by:

A. Increasing model depth
B. Capturing different relational patterns
C. Reducing token count
D. Replacing embeddings
Answer: B
Explanation: Each head learns a different aspect of meaning/order .

---

10. Encoder-only models like BERT are best for:

A. Long text generation
B. Classification and understanding
C. Translation
D. Image captioning
Answer: B
Explanation: Encoders build rich representations of input text .

---

11. A language model defines:

A. Topic clusters
B. Probability distribution over token sequences
C. Tree-based parsing rules
D. Compression of text
Answer: B
Explanation: LMs predict next token probability autoregressively .

---

12. Self-supervised training works because:

A. Labels are provided for free from the text itself
B. Humans annotate tokens
C. Only images are needed
D. It uses reinforcement learning
Answer: A
Explanation: Every sentence naturally contains missing-token prediction tasks .

---

13. A hallucination occurs when the LLM:

A. Repeats the context
B. Uses prior knowledge when context forbids it
C. Rejects answering
D. Uses transformer heads incorrectly
Answer: B
Explanation: The Shakespeare example shows prior knowledge overriding context .

---

14. Why can't we just send 1M tokens into a prompt for all tasks?

A. Models cannot embed large text
B. Extremely expensive and slow
C. Reduces context window
D. Reduces accuracy of short prompts
Answer: B
Explanation: Large context is financially and computationally unscalable .

---

15. The "Lost in the Middle" effect means LLMs:

A. Prefer long documents
B. Ignore information located in the center of prompts
C. Forget embeddings
D. Only read the first 500 tokens
Answer: B
Explanation: Attention curve is U-shaped—start and end get priority .

---

16. Why does RAG improve access control?

A. It encrypts chunks
B. Retriever only fetches allowed documents
C. Tokens are masked
D. Models store permissions internally
Answer: B
Explanation: Forbidden chunks are never retrieved → never shown to LLM .

---

17. In RAG, query embeddings are needed to:

A. Compress the document
B. Convert text into vectors for search
C. Train the LLM
D. Reduce token length
Answer: B
Explanation: Embedding → vector → nearest neighbor retrieval .

---

18. Why must we chunk documents before embedding?

A. GPU limits
B. Big documents cannot be embedded meaningfully
C. Embeddings only work on numbers
D. RNNs cannot handle long text
Answer: B
Explanation: Chunks need a Goldilocks size—precise but meaningful .

---

19. Fixed-size chunking is “dumb” because:

A. It uses too many tokens
B. It cuts sentences unnaturally
C. It compresses important text
D. It mixes embeddings

Answer: B
Explanation: Splitting by token count ignores natural boundaries .

---

20. Recursive chunking improves chunk quality by:

A. Always using paragraphs
B. Trying multiple splitting rules in order
C. Compressing text
D. Removing overlap
Answer: B
Explanation: It splits by paragraph → sentence → space until size fits .

---

21. Vector databases store:

A. Raw documents
B. Token sequences
C. High-dimensional embeddings
D. Model weights
Answer: C
Explanation: Vector DB = "library of numbers" storing chunk vectors .

---

22. RAG vs Fine-Tuning: RAG is better for:

A. Learning writing style
B. Adding new factual knowledge
C. Reducing model size
D. Tokenization
Answer: B
Explanation: RAG tells model *WHAT* to say; fine-tuning teaches *HOW* to say it .

---

23. HyDE improves retrieval by:

A. Summarizing documents
B. Generating a fake answer and embedding it

C. Expanding top-k results
D. Compressing chunks
Answer: B
Explanation: Fake answer → dense embedding → better retrieval .

---

24. Hybrid Search combines:

A. Two dense searches
B. Dense semantic + sparse keyword (BM25)
C. Sentence embeddings + RNN
D. QKV attention
Answer: B
Explanation: Semantic understands meaning; BM25 handles exact terms like IDs .

---

25. Re-ranking is necessary because:

A. Embeddings require it
B. Retriever returns too many noisy candidates
C. BM25 is slow
D. RAG requires fixed-size input
Answer: B
Explanation: Cross-encoder re-ranker selects the best among many possible chunks .

---

26. Prompt compression models remove:

A. Key concepts
B. Irrelevant filler text
C. Token embeddings
D. All punctuation
Answer: B
Explanation: They remove noise to reduce confusion during generation .

---

27. In ML systems, “upstream data” refers to:

A. Predictions
B. Features coming from external databases
C. Model outputs
D. Evaluation metrics
Answer: B
Explanation: Upstream = data flowing **into** the model pipeline (CRM, DB).

---

28. “Downstream data” refers to:

A. Logs
B. What the model sends forward to other systems
C. External API calls
D. Missing data
Answer: B
Explanation: Downstream = where predictions go after generation.

---

29. A model degrading because users move from offline → online represents:

A. Data drift
B. Concept drift
C. Prediction drift
D. Token drift
Answer: A
Explanation: Input distribution changed even though meaning of the target didn't.

---

30. Concept drift occurs when:

A. The target meaning changes
B. Users shift platforms
C. Predictions become noisy
D. Batch size changes
Answer: A
Explanation: Relationship between features and target changes over time.

---

31. OOV (Out-of-Vocabulary) hurts models like TF-IDF because:

A. They cannot tokenize numbers
B. They require fixed known words
C. They remove punctuation
D. They use deep embeddings
Answer: B
Explanation: TF-IDF can't handle unseen slang (e.g., Gen Z TikTok words).

---

32. Transformers handle OOV better because:

A. They store all words
B. They tokenize into subwords
C. They remove new tokens
D. They use BM25
Answer: B
Explanation: Subwords allow understanding even of brand-new words.

---

33. Prediction drift indicates:

A. Wrong model architecture
B. Data/concept drift upstream
C. Training failure
D. GPU overheating
Answer: B
Explanation: When predictions shift from actual distribution, something upstream changed.

---

34. A fraud model going from 5–10% predicted fraud → 50% indicates:

A. Improved accuracy
B. Severe prediction drift
C. Low recall
D. Good ROC

Answer: B
Explanation: The distribution shift signals model breakdown and need for retraining.

---

35. The correct response when a model degrades is to:

A. Increase tokens
B. Retrain immediately
C. Add noise
D. Expand context window
Answer: B
Explanation: Model drift = retraining required in any production system.

---

36. In Afiniti-style pairing models, changing group distributions requires:

A. New embeddings
B. Re-ranking
C. Retraining
D. Larger batch size
Answer: C
Explanation: When cohorts/agents shift, the mapping must relearn.

---

37. RV percentile in customer-agent ranking measures:

A. Raw accuracy
B. Ranking stability
C. How well groups and agents match statistically
D. Drift rate
Answer: C
Explanation: RV percentile reflects matching strength between customers and agents.

---

38. Why is “garbage in → garbage out” important in RAG?

A. BM25 fixes all errors
B. Irrelevant chunks mislead the LLM
C. Tokenization becomes slower
D. Embeddings get corrupted
Answer: B
Explanation: Low-quality retrieval produces hallucination-like outputs.

---

39. Which step in RAG MOST reduces hallucination?

A. Summarization
B. Re-ranking
C. QKV attention
D. Prompt compression
Answer: B
Explanation: Re-ranking ensures only the MOST relevant chunks reach the LLM.

---

40. Why does HyDE sometimes outperform direct question search?

A. The fake answer contains more semantic detail
B. It removes keywords
C. It compresses chunks
D. It avoids embeddings
Answer: A
Explanation: Dense synthetic answers retrieve better-matched documents.

1. Which scenario BEST illustrates concept drift?

A. Model predicts lower accuracy after a software update
B. People stop using cash and shift to digital payments
C. The meaning of “fraud” changes due to new regulations
D. Data pipeline breaks
Answer: C
Explanation: Concept drift = the *definition* or meaning of the target changes.

---

2. Which type of drift would occur if age distribution suddenly shifts younger?

A. Prediction drift
B. Data drift
C. Privacy drift
D. Concept drift
Answer: B
Explanation: Data drift = input feature distribution changes.

---

3. Why does a fraud model suddenly predicting 60% fraud indicate drift?

A. Too many features
B. Prediction distribution shifted
C. More GPU usage
D. TF-IDF failure
Answer: B
Explanation: Prediction drift = outputs no longer match expected distribution.

---

4. Why is RAG useful when training data becomes stale?

A. It retrains the model automatically
B. It bypasses the model and retrieves fresh external knowledge
C. It rewrites embeddings
D. It updates weights

Answer: B
Explanation: RAG uses *external documents*, avoiding reliance on old training data.

---

5. A vector database returns wrong chunks because the query is too short. What technique helps?

A. Token masking
B. HyDE
C. Recursive chunking
D. BM25-only search
Answer: B
Explanation: HyDE generates a dense fake answer that retrieves better chunks.

---

6. Why is BM25 essential in hybrid search?

A. It handles synonyms
B. It embeds numeric IDs
C. It captures exact keyword matches
D. It removes stopwords
Answer: C
Explanation: BM25 excels at precise term-based retrieval, especially IDs.

---

7. Which RAG stage directly fights "lost in the middle"?

A. Query embedding
B. Re-ranking
C. Hybrid search
D. Chunking
Answer: D
Explanation: Proper chunking ensures relevant info isn't buried in long text.

---

8. Which of these is a sign of upstream data drift?

A. Model uses too much GPU
B. CRM system changes customer field encoding
C. Predictions spike
D. Inference slows down
Answer: B
Explanation: Upstream = data entering model changes (encoding, formats, categories).

---

9. Why do TF-IDF models fail with TikTok Gen-Z slang?

A. They cannot tokenize
B. They require fixed vocabulary seen in training
C. They apply stemming incorrectly
D. They use subwords
Answer: B
Explanation: OOV (out-of-vocab) words break TF-IDF since vocab is fixed.

---

10. Transformers handle slang better because:

A. They store all words
B. Their vocabulary updates automatically
C. Subword tokenization captures unknown words
D. They use BM25 internally
Answer: C
Explanation: Subwords allow breaking slang into known pieces.

---

11. Which problem does prompt compression solve?

A. Context window overflow
B. Slow training
C. Tokenization conflicts
D. Query ambiguity
Answer: A
Explanation: Compression removes irrelevant text to fit context limits.

---

12. Why can't we rely solely on model prior knowledge for factual tasks?

A. Models forget during training
B. Prior knowledge becomes outdated
C. Models cannot store large datasets
D. Hallucinations are rare
Answer: B
Explanation: Real-world facts update faster than static LLM training data.

---

13. Which situation demands immediate model retraining?

A. A single wrong prediction
B. Slight memory increase
C. Distribution shift detected in predictions
D. New competitor product
Answer: C
Explanation: Prediction drift is the first early warning signal.

---

14. Why do we overlap chunks during fixed-size chunking?

A. Improve BM25 accuracy
B. Prevent semantic breakage between splits
C. Reduce embedding cost
D. Improve tokenization
Answer: B
Explanation: Overlap ensures sentences aren't cut abruptly.

---

15. In RAG, “Top-K retrieval” means:

A. Returning the K shortest chunks
B. Returning K most semantically similar chunks
C. Merging K vectors into one

D. Selecting K random documents
Answer: B
Explanation: Top-K = highest similarity scores in vector space.

---

16. Why is context window NOT a replacement for RAG?

A. LLMs dislike long text
B. Lost-in-the-middle reduces accuracy
C. Models cannot read PDFs
D. Tokenization fails above 500k tokens
Answer: B
Explanation: LLMs ignore the middle of huge prompts + cost issues.

---

17. What does "garbage in → garbage out" mean in retrieval?

A. RAG ignores wrong chunks
B. Irrelevant chunks cause hallucinations
C. Model retries queries
D. BM25 fixes errors automatically
Answer: B
Explanation: Bad retrieval leads to bad answers.

---

18. What causes concept drift in the Afiniti example?

A. Agent salary changes
B. Customer cohorts change behavior
C. Data missing from SQL
D. GPU overload
Answer: B
Explanation: The relationship between customer group and agent changes → concept drift.

---

19. What does re-ranking primarily optimize?

A. Sorting by timestamp
B. Precision of retrieved chunks
C. Token count
D. QKV attention quality
Answer: B
Explanation: Cross-encoder evaluates true relevance between chunk and query.

---

20. Why is RAG an “open-book exam”?

A. The model memorizes the book
B. It retrieves relevant pages and answers from them
C. It replaces embeddings
D. It improves GPU performance
Answer: B
Explanation: RAG explicitly reads external documents to answer queries.

1. Which scenario MOST accurately demonstrates *concept drift* hidden behind stable input features?

A. User age shifts from 30→25
B. Fraud definition changes due to new banking laws
C. CRM adds new columns
D. Prediction distribution shifts from 10%→50%
Answer: B
Explanation: Concept drift = target meaning changes even if input features look the same.

---

2. A model receives the same input distribution, but outputs shift dramatically. What drift is this?

A. Data drift
B. Concept drift
C. Prediction drift
D. Structural drift
Answer: C
Explanation: Prediction drift = outputs shift even if inputs look unchanged.

---

3. Which failure CANNOT be solved by RAG alone?

A. Outdated factual knowledge
B. Retrieval of irrelevant chunks
C. Misaligned tone/style
D. Missing access control
Answer: C
Explanation: Tone/style require **fine-tuning**, not RAG.

---

4. Which chunking method BEST prevents the "lost-in-the-middle" effect?

A. No chunking
B. Fixed token chunking with no overlap
C. Recursive chunking with semantic boundaries

D. BM25 re-ranking
Answer: C
Explanation: Recursive chunking ensures meaningful boundaries so content won't get buried.

---

5. Why does HyDE outperform pure semantic search for ambiguous queries?

A. It increases context window size
B. It forces the model to ignore rare words
C. The hallucinated answer is semantically denser than the question
D. It replaces embeddings with raw text
Answer: C
Explanation: Fake answers carry more meaning cues than short vague questions.

---

6. Which situation MOST likely causes *upstream* drift?

A. New user signup form adds a "preferred language" field
B. Model predicts 3× more fraud
C. GPU utilization spikes
D. Hybrid search returns fewer documents
Answer: A
Explanation: Upstream drift occurs when *data entering the pipeline* changes shape or encoding.

---

7. Why are transformers less sensitive to OOV than TF-IDF?

A. They ignore rare words
B. They retrain themselves while running
C. Subword units ensure any new word can be decomposed
D. They maintain larger dictionaries
Answer: C
Explanation: Subword tokenization eliminates OOV completely.

---

8. Which component of attention directly influences *how much* one token affects another?

A. Value vector
B. Query–Key dot product
C. Softmax temperature
D. Positional encoding
Answer: B
Explanation: Q·K determines raw attention strength before softmax.

---

9. In a vector database, which failure MOST severely harms retrieval accuracy?

A. Low-dimensional vectors
B. Using cosine similarity instead of dot product
C. Poor chunk boundaries
D. Using too many shards
Answer: C
Explanation: Even perfect embeddings fail if chunks themselves contain broken context.

---

10. Why is "garbage in → garbage out" more dangerous in RAG than in LLM-only setups?

A. LLMs ignore context
B. RAG forces the model to trust retrieved text
C. BM25 ranks garbage high
D. Embeddings become corrupted
Answer: B
Explanation: RAG *grounds* the model in whatever you retrieve—even if wrong.

---

11. Which failure MOST likely indicates *both* data drift and upstream encoding drift?

A. Online → offline sales ratio shifts
B. Feature “store_type” changes from {A,B} to {A1,B1}
C. Predictions shift suddenly
D. Re-ranking fails
Answer: B
Explanation: Encodings being altered upstream change categorical meaning entirely.

---

12. Why is a 1M-token window STILL inferior to RAG even when cost is ignored?

A. Attention is nonlinear
B. Token embeddings degrade at high temperatures
C. LLMs overweight the beginning and end, ignoring the middle
D. Position encodings break after 100k tokens
Answer: C
Explanation: U-shaped attention curve → “lost-in-the-middle”.

---

13. Why does re-ranking often outperform increasing Top-K retrieval size?

A. Re-ranking reduces embedding size
B. Top-K only improves recall, but re-ranking improves precision
C. Re-ranking modifies embeddings
D. Top-K doesn’t work with semantic search
Answer: B
Explanation: Large Top-K brings noise; re-ranking filters noise.

---

14. Which retrieval failure can HyDE NOT fix?

A. Short ambiguous queries
B. Misspelled queries
C. Poor chunk segmentation
D. Missing factual documents
Answer: D
Explanation: HyDE retrieves better, but cannot retrieve documents that don’t exist.

---

15. In the Afiniti example, why does ranking degrade when agents' availabilities shift?

A. Data drift
B. Concept drift
C. Prediction drift
D. Feature leakage
Answer: B
Explanation: Relationship between customer cohorts and agents changes → concept drift.

---

16. Which is the strongest signal that you MUST retrain immediately?

A. Context window overflow
B. Rare words appear
C. Prediction distribution diverges from actual labels
D. Lower GPU usage
Answer: C
Explanation: Prediction drift = early strong signal of model degradation.

---

17. What is the core mathematical reason RAG reduces hallucination?

A. It adjusts internal weights
B. It constrains the model's conditional probability distribution to retrieved facts
C. It increases embeddings
D. It replaces softmax
Answer: B
Explanation: Conditioning on retrieval reduces the entropy of the output distribution.

---

18. Which step makes hybrid search robust to semantic and lexical mismatches?

A. Cross-encoder
B. BM25 + dense embeddings combined
C. Recursive chunking
D. Fine-tuning encoder
Answer: B
Explanation: Dense handles meaning; BM25 handles exact words/IDs.

---

19. Why does adding *more* retrieved chunks sometimes decrease accuracy?

A. Transformer heads saturate
B. Softmax overfits
C. Extra chunks introduce noise that misleads attention
D. Key vectors explode
Answer: C
Explanation: Irrelevant chunks dilute the attention signal.

---

20. Why are embeddings considered “high-dimensional meaning coordinates”?

A. They store syntax trees
B. Every coordinate corresponds to a semantic direction
C. They represent tokens as integers
D. They include model weights
Answer: B
Explanation: Each dimension captures latent properties—gender, tense, topic, etc.